Marmara Üniversitesi
Electrical and Electronics Engineering Department
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EE 3016/3051 – Quiz 1
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Vectors			
0	$\vec{D} = \vec{a}_x - (x+y)\,\vec{a}_y + \vec{a}_z$	5	$\vec{D} = \vec{a}_x - (2x+y)\,\vec{a}_y + y\,\vec{a}_z$
1	$\vec{D} = xy\,\vec{a}_x - y^2\,\vec{a}_y + \vec{a}_z$	6	$\vec{D} = xy\,\vec{a}_x - z^2\,\vec{a}_y + \vec{a}_z$
2	$\vec{D} = y\,\vec{a}_x - (y+z)\,\vec{a}_y + 2\,\vec{a}_z$	7	$\vec{D} = y\,\vec{a}_x - (x+z)\,\vec{a}_y + 2z\,\vec{a}_z$
3	$\vec{D} = (x+z)\,\vec{a}_x + y\,\vec{a}_y + 2x\,\vec{a}_z$	8	$\vec{D} = (x+y)\,\vec{a}_x + z\,\vec{a}_y + 2x\,\vec{a}_z$
4	$\vec{D} = 2y^2\,\vec{a}_x + \vec{a}_y + x^2\,\vec{a}_z$	9	$\vec{D} = 2x^2\,\vec{a}_x + \vec{a}_y + z^2\,\vec{a}_z$

Points			
0	$P(0,1,2)$	5	$P(-1,0,2)$
1	$P(1,0,2)$	6	$P(0,1,-2)$
2	$P(-1,1,0)$	7	$P(2,1,0)$
3	$P(1,0,-2)$	8	$P(2,0,-1)$
4	$P(0,0,-2)$	9	$P(0,2,2)$

Questions

1) Find the total flux of the field $\vec{D}$ over the surface of the given structure. Show all work

Structure defined by: $1 < x < 3, \quad 0 < y < 1, \quad -1 < z < 1$

2) Find the electric field intensity due to the finite line charge at the point P. Assume a uniform line charge density of 1 mC/m. Show all work.

The finite line is defined between points: (0, 0, 0) and (0, 1, 0)

$$\vec{E}_p = \frac{-\rho_L}{4\pi\varepsilon\, s}\left[(\sin\alpha_2 - \sin\alpha_1)\,\vec{a}_n - (\cos\alpha_2 - \cos\alpha_1)\,\vec{a}_L\right]$$